

IDENTIFY WHICH OF THE FOLLOWING ARE SETS.

1. Collection of all months of a year beginning with letter J.
2. The collection of ten most talented writers in India.
3. A team of eleven best cricket batsmen in India.
4. The collection of all boys in your class.
5. The collection of all natural numbers less than 100.
6. The collection of most dangerous animals in the world.
7. The collection of all short students in your class.
8. The collection of all even integers less than 25.

REPRESENT THE FOLLOWING SETS IN ROSTER FORM

1. $A = \{x: x \text{ is a natural number less than } 6\}$
2. $B = \{x: x \text{ is an integer and } -3 \leq x < 7\}$
3. $C = \{x: x \text{ is a two-digit number such that the sum of its digits is } 8\}$
4. $D = \{\text{The set of all letters in the word TRIGNOMETRY}\}$
5. $E = \{\text{The set of all the letters in the word BETTER}\}$
6. $F = \{x: x \text{ is a prime number which is a divisor of } 60\}$
7. $G = \{x: x \in \mathbb{R}, 3 \leq x \leq 4\}$

REPRESENT THE FOLLOWING IN A SET BUILDER FORM

1. $A = \{a, e, i, o, u\}$
2. $B = \{0, 3, 6, 9, 12, 15, 18, \dots\}$
3. $C = \{1, 1/2, 1/3, 1/4, 1/5, \dots\}$
4. $D = \{1, 1/4, 1/9, 1/16, 1/25, \dots\}$
5. $E = \{1/2, 2/3, 3/4, 4/5, 5/6, 6/7, 7/8, 8/9, 9/10\}$
6. $F = (6, 12)$
7. $G = (-5, 9]$

MATCH THE FOLLOWING

(i) $\{F, L, O, W\}$	(a) $\{x: x \text{ is a positive integer and is divisor of } 18\}$
(ii) $\{1, 4, 9, 16, 25, 36\}$	(b) $\{x: x \text{ is a prime number between } 40 \text{ and } 50\}$
(iii) $\{1, 2, 3, 6, 9, 18\}$	(c) $\{x: x = n^3, n \in \mathbb{N} \text{ and } n^3 < 80\}$
(iv) $\{41, 43, 47\}$	(d) $\{x: x \text{ is a letter of the word FOLLOW}\}$
(v) $\{1, 8, 27, 64\}$	(e) $\{x: x = n^2, n \in \mathbb{N} \text{ and } n^2 < 40\}$

DETERMINE THE CARDINALITY OF THE FOLLOWING SETS

1. $A = \{1, 3, 9, 12, 15, 18, 24\}$
2. $B = \{0, 2, 4, \dots, 20\}$
3. $C = \{\}$
5. $D = \{5, 5, 5, 5, 5\}$
6. $E = \{R, E, M, E, M, B, E, R\}$
7. $F = \{\emptyset\}$
8. $G = \{\emptyset, \{\emptyset\}\}$
8. $H = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$

WHICH OF THE FOLLOWING ARE EMPTY SETS?

1. $A =$ set of odd natural numbers divisible by 2
2. $B =$ set of even prime numbers.
3. $C = \{x: x \text{ is a natural number, } x < 5 \text{ and } x > 6\}$
4. $D = \{x: x \text{ is a point common to any 2 parallel lines}\}$
5. $E = \{x: x \text{ is a real number and } x^2 \text{ is } -1\}$

SUBSETS AND POWERSET.

1. List all the subsets of $\{a, b, c\}$
2. List the proper subsets of $A = \{1, 2, 3\}$
3. Given A is the set of letters in the word MOON. Find the power set of A .
4. Given $A = \{0\}$, Find the power set of A .
5. Find the power set of $\{\{a, b\}, c\}$
6. Find the power set of $A = \{1, 2, \{3, 4\}\}$
7. Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 3\}$ and $C = \{2, 4\}$

Find all sets of X such that (i) $X \subset P(B)$ and $X \subset P(C)$

COMPLEMENT

1. Find the complement of P for the given conditions.

$$P = \{x: x \text{ is a digit } > 3 \text{ and } < 10\}$$

$$U = \{x: x \text{ is a positive number smaller than } 10\}$$

2. If $U = \{1,2,3,4,5,6,7,8,9\}$ find the complements of the following sets
(i) $A = \{2,4,6,8\}$ (ii) $B = \{1,3,5,7,9\}$ (iii) $C = \{2,3,5,7\}$ (iv) $\emptyset$ (v) U

OPERATIONS ON SETS

1. Let $A = \{1,2,3,4\}$, $B = \{2,3,6,7,9\}$ find $A \cup B$.
2. If $A = O$ (set of odd natural numbers), $B = E$ (Set of even natural numbers) then find $A \cup B$.
3. If $A = \{\text{set of rational numbers}\}$ and $B = \{\text{set of irrational numbers}\}$, then find $A \cup B$.
4. If $A = \{x: x^2 - 4, x \in I\}$ and $B = \{y: y^2 - 9, y \in I\}$ find $A \cup B$
5. If $A = \{x: 1 < x < 5, x \in \mathbb{N}\}$ and $B = \{y: 3 < y < 7, y \in \mathbb{N}\}$ find $A \cup B$.
6. If $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{5, 6, 7, 8\}$. Find $A \cup B \cup C$.
6. If A is a set of odd natural numbers and B is a set of even natural numbers, then find $A \cap B$.
7. If set $M = \{x \mid x \text{ is a prime number less than } 20\}$ and set $N = \{x \mid x \text{ is an odd number less than } 10\}$, what is $M \cap N$?
8. If $A = \{3, 5, 7, 9, 11\}$, $B = \{7, 9, 11, 13\}$, $C = \{11, 13, 15\}$. Find $A \cap (B \cup C)$.
9. If $A = \{1,2,3,4\}$ and $B = \{2,4,7,9\}$ then find $A - B$ and $B - A$
10. If $A = \{12,15,17,20,21\}$, $B = \{12,14,16,18,21\}$ and $C = \{15,17,18,22\}$ then find $A - B$, $B - C$, $C - A$, $B - A$, $A - A$
11. For sets $A = \{x \mid x \text{ is an integer, } 1 \leq x \leq 6\}$ and $B = \{x \mid x \text{ is an even integer, } 2 \leq x \leq 8\}$, find the set $A - B$.
12. If set $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$ and $C = \{0, 2, 4, 6, 8\}$. Then write the universal set for all three sets.

13. If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $A = \{1, 3, 5, 7, 9\}$. Find A' .
14. Given the sets $A = \{10, 15, 17, 19, 20\}$ and $B = \{15, 16, 18\}$. Find out the difference between both sets A and B and also find out the symmetric difference between them.
15. Given sets $X = \{a, b, c, d\}$ and $Y = \{b, d, e, f\}$, find the symmetric difference of X and Y (denoted as $X \Delta Y$).
16. Given $U = \{x: x \text{ is a natural number}\}$, $B = \{2x: x \in U\}$, $C = \{2x+1: x \in U\}$
Find (i) $B \cup C$ (ii) $B \cap C$ (iii) $U - C$ (iv) B'